

Asymmetric Voter Turnout Across Elections

Strategies for Equitable Civic Participation in Quincy, MA and across Municipal Districts

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This project develops a data-driven classification framework to identify residents of Quincy, Massachusetts who are unlikely to vote in municipal elections and may therefore be appropriate candidates for targeted civic outreach. With insight into the local political involvement trends of Quincy given by client Dr. Russ Wilcox, the analysis focuses on the sharp gap between federal and local election participation, especially among voters who participate in high-salience national elections but do not consistently turn out for city-level contests. This project uses voter file data from L2 Data, precinct-level municipal election turnout information, and contextual information about local issues in Quincy through 2025, to analyze which voters participated in Quincy’s 2023 municipal election cycle, which precincts had higher turnout rates than others, and what demographic, geographic, and behavioral factors are associated with increased local participation.

The central research question is: *How can Quincy better incentivize voter turnout for the 2027 mayoral election, given evidence of higher turnout in previous local elections?*

I. INTRODUCTION

Municipal elections often shape the policies that most directly affect residents’ daily lives, yet they consistently receive far less voter participation than federal elections. In cities like Quincy, Massachusetts, this gap creates a major civic problem: the people most affected by local decisions may not be the people most likely to vote in local races. Federal elections tend to attract a broad electorate, while municipal elections are often dominated by older, wealthier, and more consistent voters. On average, local or municipal elections receive 30.1% fewer votes in comparison to Presidential elections [8]. As a result, city leadership, school policy, public services, and community priorities may be shaped by a relatively narrow portion of the population.

Quincy reflects this broader local-turnout challenge. Approximately 30% of registered voters participated in the 2023 city election, leaving most eligible residents unrepresented in decisions concerning the mayor, City Council, and School Committee [12]. A precinct-level analysis by QuincyVotes found that homeownership and racial composition were strongly associated with 2023 turnout differences across Quincy precincts [13]. The organization reported that these two characteristics together accounted for 84% of the observed precinct-level variation, although this ecological association should not

be interpreted as proof that either characteristic directly caused individual turnout.

To address this problem, we developed a classification model to identify low-propensity municipal voters in Quincy and provide statistical insights into methods that may increase turnout in future local elections. Our client, Dr. Russ Wilcox, is particularly interested in the difference between voters who participate in federal elections and those who turn out for municipal contests. This distinction is important because many residents are politically engaged at the national level but remain absent from local elections, even though local government decisions often have a more immediate impact on schools, neighborhoods, infrastructure, and community representation.

The project is notably motivated by turnout patterns surrounding Quincy’s recent municipal elections, especially the 2025 local election and locally salient education issues. According to Dr. Wilcox, turnout in 2025 was higher than anticipated, raising important questions about which precincts participated in the elections the most, which voters participated for the first time in a local election, and what local issues may have motivated residents to vote. One locally salient example was the debate over recognizing Lunar New Year as an official Quincy Public Schools holiday. Students and community members advocated for the policy beginning before

the School Committee’s May 2023 vote, emphasizing its importance to Quincy’s Asian American families. At the time, Asian students represented approximately 39% of the district and formed its largest racial group [3]. The issue remained active in 2025, when the School Committee rejected the proposal for the third time since May 2023. Nearly 40% of Quincy Public Schools students identified as Asian American, and more than 20% of students were absent citywide on Lunar New Year in January 2025 [11]. Community testimony frequently framed the issue in terms of cultural recognition and representation. This is an example of how local policy issues may increase the perceived relevance of a municipal election, suggesting that turnout is not only a function of voter demographics, but also of whether residents feel that a municipal election is connected to issues that matter to their community.

For this analysis, our dataset does not cover the 2025 local election, which garnered $\sim 23\%$ voter turnout [2]. The latest local election included in our dataset occurred in 2023. However, the 2023 local election, like the 2025 election, showed relatively high turnout compared with earlier municipal election cycles, as shown in Figure 1.

According to Dr. Wilcox, another salient issue is how local and state elections shape broader debates over digital access, social media regulation, and free speech. This is especially relevant in Massachusetts, where recent state-level proposals concerning youth social media use, age verification, school phone restrictions, and parental access to minors’ online data have raised concerns from civil liberties and privacy advocates about surveillance, excessive regulation, and access to online information. This is another specific case of why turnout discrepancies are a concern: when residents do not participate in municipal, state representative, or state senate elections, they may have less influence over policies that determine not only schools and local services, but also digital privacy, speech rights, and access to online platforms. For this reason, the project treats voter turnout as a civic access issue, where low participation can affect whose voices are represented in decisions about both physical communities and digital public life.

Using L2 voter file data, precinct-level election turnout trends, and structured information about local issues, this analysis examines who participates in Quincy municipal elections, who does not, and how turnout rates differ across precincts and demographic groups. Official Quincy election results are published by the Office of the City Clerk, which organizes results by election date, from 2005 to 2025 [5]. Our objective involves identifying “low-propensity” and “federal-only” voters: residents who vote infrequently in municipal elections or only vote in presidential or other national elections while forgoing municipal contests entirely. For this study, federal-only voters are specifically defined as individuals who participated in at least one general or presidential election but had no recorded municipal-election participation, serving as a proxy for voters who are engaged in high-salience

elections but inactive locally. This project accounts for data quality challenges in the voter file, including inconsistent turnout indicators and blank municipal election fields.

This analysis aims not merely to predict which voters are unlikely to participate but ultimately to make the prediction useful for civic outreach. By identifying low-propensity municipal voters and the factors associated with their lack of participation, the final predictive model can help Quincy voter participation advocacy programs design more effective turnout strategies for the 2027 mayoral election. These strategies may include precinct-level targeting, issue-based messaging, and outreach to demographic groups that are underrepresented in local elections. This project intends to provide a practical framework for understanding how Quincy can broaden participation in municipal democracy.

II. DATASET

The dataset used for this project comes from L2 Data, a national voter file vendor that provides public voter roll information combined with modeled demographic, geographic, consumer, and political behavior variables [9]. The original dataset contained 33,315 voter records and 900 columns. No duplicate voter rows were found. Before the analysis and after consulting with Dr. Wilcox, the dataset was first cleaned by removing 294 completely empty columns, leaving 33,315 rows and 606 retained variables. These removed columns corresponded largely to jurisdiction-specific fields that were not applicable across all cities, towns, or districts. In some cases, L2 Data was unable to populate the fields because the relevant political or administrative unit did not exist for that jurisdiction, or because there were no observations associated with that category. For example, the water district column was empty for Quincy because Quincy does not operate through a separate water district system in the same way that some other municipalities or special districts do. As a result, these fields were treated as structurally missing rather than as meaningful voter-level information. No voter records were removed during this initial cleaning step.

After this initial cleaning, additional pre-processing was conducted to create a model-ready version of the dataset. We removed variables with extremely high missingness ($> 80\%$ missingness), constant or near-constant variables, raw address and personally identifying fields, raw precinct-level election return variables, and other columns that were unlikely to provide useful or appropriate predictive information. This reduced the final processed dataset to 33,315 voter records and 292 columns. The goal of this reduction was not only to make the dataset smaller, but also to make the modeling process more interpretable, reduce noise, and avoid using variables that could introduce privacy concerns or data leakage. Because the model uses voter-file and vendor-

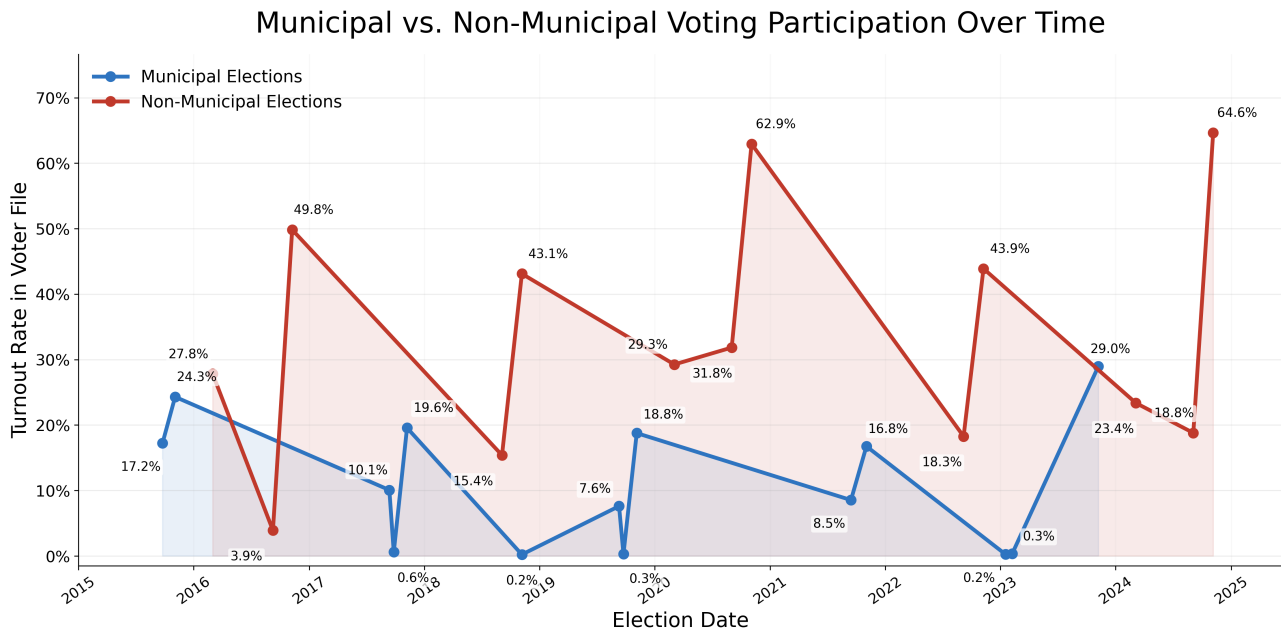


FIG. 1. **Municipal and non-municipal voting participation over time.** This figure shows the share of voters in the cleaned voter file who participated in Quincy municipal and non-municipal elections. Participation fluctuates substantially across election cycles, with lower rates in some off-cycle municipal elections and a notable increase to 29.0% in the most recent election shown in 2023. This trend highlights the uneven nature of local election engagement and helps identify periods when local issues or campaign activity may have successfully mobilized more voters. The plotted elections differ in type and salience, so adjacent points are not necessarily directly comparable.

modeled demographic data, any operational use should follow privacy-preserving practices and avoid targeting individuals with sensitive or exclusionary messaging.

Each row in the dataset represents an individual voter. The remaining columns include several broad categories of information:

- 1. **Voter registration and party information**
These include fields such as registration status, registration date, party description, and voter activity status. Registration dates were converted into date variables and used to calculate registration year and years registered. Party registration was retained because it provides a useful, interpretable measure of political affiliation or non-affiliation.
- 2. **Geographic and household information**
The dataset includes precinct, ward, district, and household-level variables. These fields are important because one goal of the project is to understand turnout differences across Quincy precincts and identify which areas may require more targeted municipal outreach. Exact addresses and other personally identifying location fields were removed, while broader geographic indicators such as precinct and ward were retained.
- 3. **Demographic variables**
The data includes variables such as gender, birthdate, household composition, and other modeled

demographic indicators. For age, we recalculated age using birthdate rather than relying only on the dataset’s static age field, because the original age variable reflects a snapshot from when the data was collected. We also corrected two-digit birth years that were incorrectly parsed into future dates, so dates such as 9/16/61 were interpreted as 1961 rather than 2061.

- 4. **Election history variables**
A major strength of the dataset is that it contains voter participation history across multiple election types, including general elections, primaries, presidential primaries, and local or municipal elections. The raw election-history columns were standardized into binary indicators, where recorded participation was coded as 1 and non-participation or blank values were coded as 0. After creating these binary versions, the original raw election columns were removed. This allowed us to preserve turnout information while making the election-history variables consistent and usable for analysis.
- 5. **Adjusted turnout measures**
Because younger and recently registered voters had fewer opportunities to vote, we created age- and registration-adjusted turnout measures. For each voter, we calculated the number of municipal and non-municipal election events for which they were eligible based on birthdate and registration date.

We then divided the number of recorded votes by the number of eligible elections to create adjusted municipal and non-municipal turnout rates. This prevents younger voters or newly registered voters from being incorrectly classified as low-propensity simply because they had fewer elections available to them.

6. Consumer and modeled behavior variables

L2 also includes consumer and modeled behavior variables, such as household characteristics, lifestyle indicators, demographic estimates, and interest-based variables. Consumer variables that appeared as “Y” or blank indicators were converted into binary variables when appropriate. Percent-style variables, including voter performance and area-level percentage fields, were converted into numeric decimal format, while values such as “Unknown” or “Not Eligible” were treated as missing rather than as zero. These variables may help explain differences in municipal turnout, but they are interpreted cautiously because vendor-modeled data can have uneven coverage and potential bias across groups.

A key data issue in this project is the interpretation of blank values. In the election history columns, turnout is often recorded with a value such as “Y” when a person voted, while blank values generally indicate no recorded vote for that election. However, because election data is not always standardized across columns, these fields were carefully converted into binary variables before modeling. Similarly, consumer data fields required additional pre-processing because some variables used “Y” or blank values, while others contained percentages, category labels, or missing-value labels such as “Unknown” and “Not Eligible.”

The final processed dataset is well-suited for the project because it allows us to distinguish between several important voter types. In particular, we can identify residents who participate in federal or general elections but not municipal elections, residents who consistently vote locally, and residents who appear to be low-propensity municipal voters even after accounting for age and registration history. These distinctions are central to our research question: *how can Quincy identify low-propensity municipal voters and develop strategies to increase turnout in the 2027 mayoral election?*

III. EXPLORATORY DATA ANALYSIS

Before building the classification model, we conducted exploratory data analysis to understand the structure of the cleaned L2 voter file and identify patterns directly relevant to Dr. Russ Wilcox’s goals. The purpose of the EDA was not simply to summarize the dataset, but to determine which voters are least engaged in municipal elections, how low-propensity voting varies across

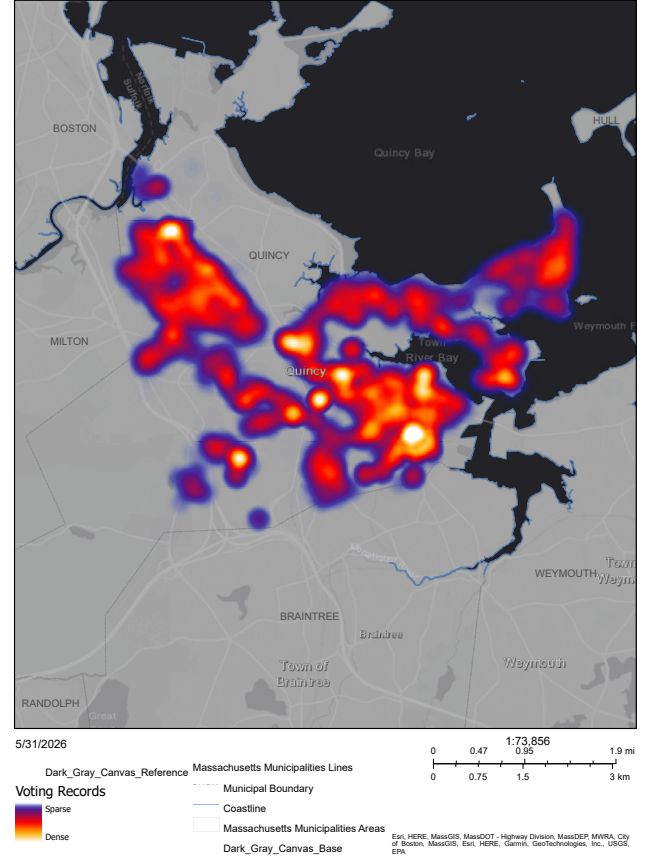


FIG. 2. Spatial distribution of voting records in Quincy. This heat map shows the geographic concentration of voter records across Quincy and nearby municipalities, with warmer colors indicating denser clusters of recorded voters and cooler colors indicating sparser areas. The map highlights that voter records are not evenly distributed across the city, with higher concentrations around central and coastal Quincy neighborhoods and lower density in surrounding or less populated areas.

precincts and party registration, and where Quincy may have the strongest opportunity for targeted outreach.

We wanted to quantify the difference between federal election participation and municipal election participation. Many voters participate in presidential or federal elections but do not vote in local races, even though municipal elections often determine school policy, local leadership, public services, and neighborhood-level priorities.

To study this gap, we created several turnout-based variables from the raw L2 election-history columns. First, we converted each recorded election column into a binary indicator, where a recorded vote was coded as 1, and no recorded vote was coded as 0. We then separated these election indicators into municipal and non-municipal elections, allowing us to calculate each voter’s total municipal votes, total general election votes, total primary votes, and total non-municipal votes.

Because not every voter was eligible for the same number of elections, we also created adjusted turnout measures. For each voter, we calculated how many municipal and non-municipal election events they were eligible for based on birthdate and registration date. We then divided the number of recorded votes by the number of eligible elections to create `municipal_turnout_rate_adjusted` and `non_municipal_turnout_rate_adjusted`. These variables supplement our target variable in our classification model of whether a registered voter voted in any of three 2023 municipal elections.

To avoid classifying younger or recently registered voters as low-propensity simply because they had fewer opportunities to vote, we define low-propensity municipal voting using an eligibility-adjusted turnout rate.

For voter i , let M_i be the number of municipal elections in which the voter participated, and let E_i be the number of municipal elections for which the voter was eligible based on age and registration date. The adjusted municipal turnout rate is defined as:

$$T_i^m = \frac{M_i}{E_i}$$

where $E_i > 0$. We then classify a voter as low-propensity if their adjusted municipal turnout rate is less than or equal to 25%:

$$LP_i = \begin{cases} 1, & \text{if } T_i^m \leq 0.25 \\ 0, & \text{if } T_i^m > 0.25 \end{cases}$$

Voters with no eligible municipal elections are not assigned a low-propensity label because their turnout rate cannot be meaningfully calculated.

From these measures, we created two main outreach variables. The first is the low-propensity municipal voter indicator, which identifies voters with weak municipal turnout after adjusting for eligibility. The second is the federal-only voter indicator, which identifies voters who participate in federal or general elections but do not consistently participate in municipal elections. This second group is especially important for Dr. Wilcox’s question because these voters are not necessarily politically disengaged; rather, they may already vote in higher-salience elections but have not been mobilized around local issues.

Missingness

The first EDA step was to evaluate missingness in the cleaned dataset, shown in Figure ???. Although 294 completely empty columns were removed before analysis, several remaining variables still had extremely high missingness. Many of the most incomplete variables were consumer-modeled fields, including candidate-preference,

issue-interest, and lifestyle variables. This is important because L2 combines public voter roll information with vendor-modeled consumer and behavioral data [9]. These modeled variables may have uneven coverage across voters, so highly sparse fields were either excluded from modeling or treated cautiously. For the classification model, more reliable variables included election history, precinct, party registration, household composition, registration date, and calculated age.

TABLE I. **Top variables by missingness in the cleaned L2 voter file.** After removing 294 completely empty columns, several remaining variables still contained partial missingness. Many of the highest-missing variables were consumer-modeled fields, suggesting that these variables should be excluded, imputed, or interpreted cautiously during modeling.

Variable	Missing Percent
ConsumerDataLL Palin Sarah	100.0%
ConsumerDataLL Bachmann Michelle	100.0%
ConsumerDataLL Cain Herman	100.0%
ConsumerDataLL McCain John	100.0%
ConsumerDataLL Church Attendee	100.0%
ConsumerDataLL Bush George	100.0%
ConsumerData African American Professional in Household	100.0%
ConsumerDataLL Clinton Hillary	100.0%
ConsumerData Donor Environmental Issues	100.0%
ConsumerDataLL Election of Conservative Judges	100.0%
ConsumerData High End Appliances	100.0%
ConsumerDataLL Romney Mitt	100.0%
ConsumerDataLL Gun Control	99.9%
ConsumerDataLL Obama Barack	99.9%
Mailing Addresses NCOA CancelDate	99.9%

Voter Patterns

Figure 1 shows municipal voting participation over time using the binary municipal election indicators. This figure provides context for the turnout problem by showing that municipal participation fluctuates substantially across election cycles. Participation is lowest in some off-cycle local elections and rises sharply in the most recent election shown. This supports the idea that local turnout is not fixed; rather, it may respond to candidate salience, local issues, community mobilization, or broader election context. This is especially relevant to Dr. Wilcox’s interest in understanding why turnout increased in recent local elections and how that knowledge could inform outreach before the 2027 mayoral election.

The precinct-level low-propensity graph in Figure 3 shows that municipal turnout behavior varies substantially across the Quincy precincts represented in the supplied voter-file extract. The share of voters classified as low-propensity ranges from approximately 48.2% in Quincy 1-4 to 82.7% in Quincy 4-1. This is a major finding because it suggests that low municipal participation is not evenly distributed across the city. Some precincts have much larger concentrations of voters who are unlikely to participate in municipal elections. This directly connects to Dr. Wilcox’s question about which

precincts "performed" best and where turnout efforts should be focused. Precincts with lower low-propensity rates may represent areas where local engagement is already stronger, while precincts with higher rates may require more direct outreach, stronger issue-based messaging, or more localized mobilization before the 2027 mayoral election.

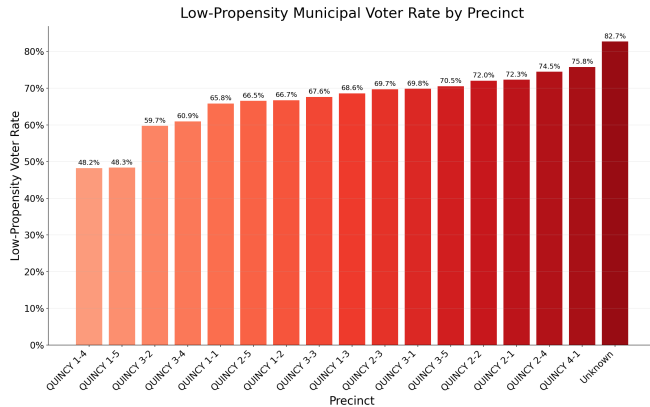


FIG. 3. Low-propensity municipal voter rate by Quincy precinct. The share of voters classified as low-propensity varies substantially across precincts, ranging from approximately 48.2% in Quincy 1-4 to 82.7% in Quincy 4-1. This variation suggests that municipal turnout challenges are not evenly distributed across the city, and that precinct-level targeting may be useful for identifying areas where voter outreach should be prioritized before the 2027 mayoral election.

The federal-only voter rate by precinct in Figure 4 provides another important perspective. This graph shows the share of voters in each precinct who vote in federal or general elections but do not consistently vote in municipal elections. Federal-only voter rates vary from approximately 26.6% in Quincy 1-4 to 53.9% in Quincy 4-1. This pattern is especially useful because federal-only voters are likely more reachable than voters with no voting history at all. They have already shown that they are willing to participate in elections, but they may not see municipal elections as personally relevant. For Dr. Wilcox's goals, these voters represent a strong outreach group: the challenge is not convincing them that voting matters in general, but showing them that local elections matter for schools, representation, city services, and neighborhood-level decisions.

Figure 5 compares each voter's total general election votes with their total municipal election votes. The positive trend line shows that voters with more general election participation also tend to have higher municipal election participation. This suggests that voting behavior is cumulative: voters who are already engaged in higher-salience elections are more likely to participate locally as well. At the same time, the cluster of voters near zero general and municipal votes shows that low turnout is also persistent across election types. This finding supports the use of prior voting behavior as an important descriptive feature in the EDA, while also motivating the

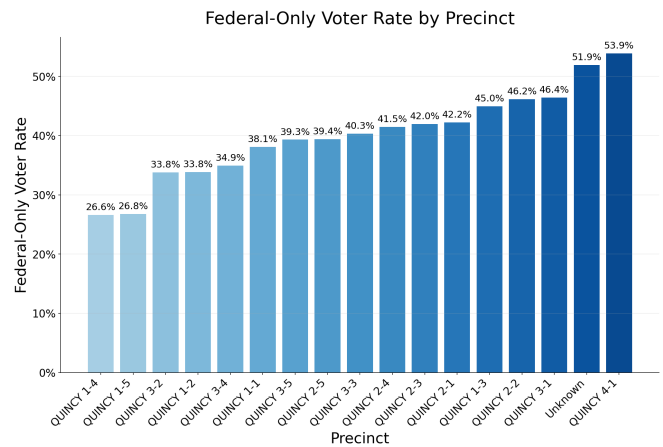


FIG. 4. Federal-only voter rate by Quincy precinct. This figure shows the share of voters in each precinct who participate in federal or general elections but do not consistently vote in municipal elections. Federal-only voter rates vary from approximately 26.6% in Quincy 1-4 to 53.9% in Quincy 4-1, suggesting that some precincts contain a much larger pool of politically engaged but locally inactive voters. These precincts may be especially important targets for outreach because residents have already demonstrated willingness to vote, but may need stronger issue-based messaging or mobilization to participate in municipal elections.

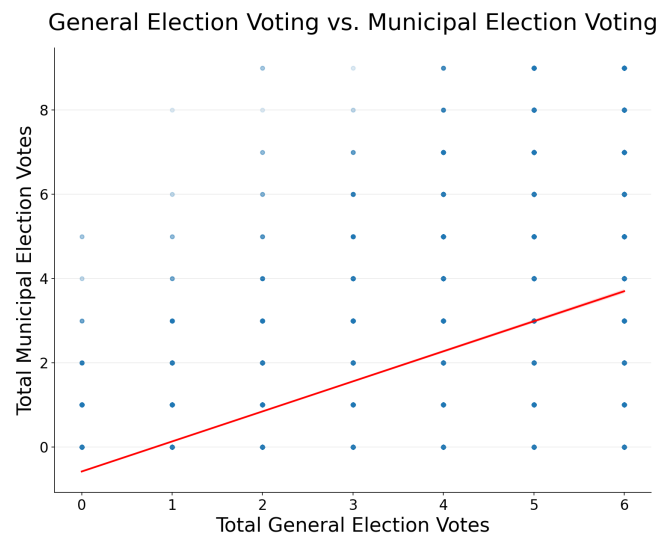


FIG. 5. General election voting versus municipal election voting. This figure compares total general election votes with total municipal election votes for voters in the cleaned Quincy voter file. The positive trend line suggests that voters who participate more frequently in general elections also tend to participate more frequently in municipal elections. Conversely, voters with little or no general election history are also more likely to have low municipal participation. This pattern highlights the relationship between broader electoral engagement and local turnout.

need to identify voters who participate federally but remain inactive in municipal elections.

Figure 6 shows that low-propensity municipal voting also differs by party registration. Democratic voters have the lowest low-propensity rate at 52.4%, followed by Republican voters at 54.6% and Other voters at 58.8%. Non-partisan voters have the highest low-propensity rate at 69.0%. This suggests that voters without a major-party affiliation may be especially disconnected from municipal elections. In a local election context, this matters because municipal races often rely less on national party identity and more on candidate familiarity, local issues, and direct voter contact. The high low-propensity rate among non-partisan voters may indicate that these voters require different outreach strategies than registered Democrats or Republicans. Rather than emphasizing party-based appeals, campaigns may need to use issue-based messaging connected to schools, local representation, public services, or community-specific concerns.

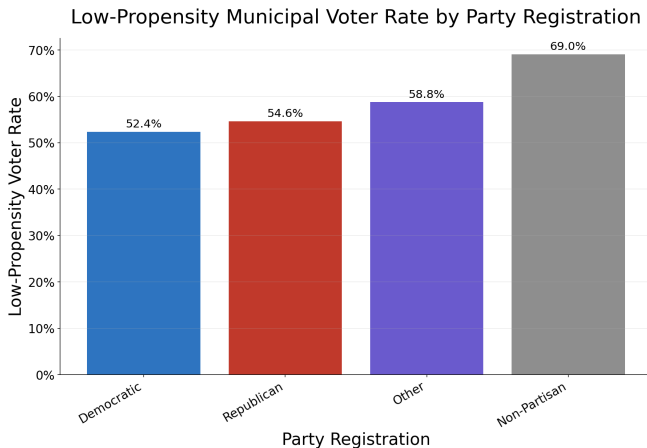


FIG. 6. **Low-propensity municipal voter rate by party registration.** This figure compares the share of voters classified as low-propensity across party registration groups. Non-partisan voters have the highest low-propensity rate at 69.0%, followed by Other voters at 58.8%, Republican voters at 54.6%, and Democratic voters at 52.4%. This suggests that voters without a major-party affiliation may be less consistently engaged in municipal elections, making them an important group for targeted outreach before the 2027 mayoral election.

Taken together, these EDA results suggest several priorities for the modeling stage. First, precinct should be important because both low-propensity municipal voting and federal-only voting vary meaningfully by geography. Second, party registration and household political context may help explain differences in municipal turnout, but they should be interpreted carefully because local elections are not always driven by the same partisan dynamics as federal elections. Third, missingness analysis shows that many consumer-modeled variables are too sparse to use reliably, so the model should prioritize more complete and interpretable variables. Finally, the

federal-only voter group should be treated as a key outreach segment because these voters are politically active in some elections but absent from municipal contests.

This EDA supports the broader purpose of the project: helping Quincy move from general turnout concerns to targeted, evidence-based outreach. Instead of assuming that all low-turnout areas or low-propensity voters are the same, the analysis shows clear differences by precinct, party registration, and election type. These differences can help the city identify where municipal disengagement is most concentrated, which voters may be most reachable, and what types of outreach may be most effective before the 2027 mayoral election.

IV. METHODS

To predict local municipal election non-participation among registered voters in Quincy, we constructed a supervised machine learning model. Our primary modeling goal was to identify voters most likely to skip the 2023 municipal election cycle. We then supplemented the main analysis with a second model that tested whether the same predictors could also identify voters with a broader history of low municipal election participation. This approach can be adapted to other election years and jurisdictions to estimate municipal non-voting risk among registered voters.

Separately from and in addition to the supplemental robustness model, we tested an expanded XGBoost specification using the primary 2023 outcome and a broader predictor set. This model included pre-2023 voting-history summaries and age- and registration-adjusted turnout measures, which account for the number of elections in which each voter was eligible to participate. The expanded model achieved a ROC AUC of 0.92, compared with approximately 0.802 for the reduced XGBoost model, indicating a modest improvement in predictive performance.

However, the expanded model relied heavily on prior voting behavior. Having any prior municipal vote accounted for 32.7% of its feature importance, and the five most important predictors were all voting-history measures, together accounting for approximately 50.3% of total importance. The model therefore primarily confirmed that voting is habitual: voters with a history of participation were more likely to vote again, while voters with limited prior participation were more likely to abstain.

Although this expanded model was useful as a predictive benchmark, the results offered us fairly redundant information, so we selected the reduced 18-predictor model for the main analysis. Its small reduction in ROC AUC was offset by clearer interpretation of demographic, socioeconomic, political, residential, and geographic correlates of non-voting. These predictors provide more useful guidance for identifying populations and areas that may benefit from outreach. The expanded model demon-

strates the importance of voting habits, while the reduced model better supports the project’s goal of developing actionable and interpretable outreach priorities. Neither specification should be interpreted causally.

The main model uses a reduced XGBoost specification. XGBoost was selected as the final project model because it achieved the strongest overall ranking performance while still using a compact and interpretable predictor set. Rather than relying on hundreds of voter-file variables, the final model uses 18 theoretically motivated predictors related to demographics, political context, socioeconomic status, household characteristics, residential stability, and geography. As a supplemental robustness check, we applied the same 18 predictors to a broader low-propensity municipal turnout target to evaluate whether the same predictors generalized beyond the 2023 election cycle.

The primary outcome variable was `DidNotVote`, a binary indicator for whether voter i failed to participate in the 2023 municipal election cycle. Let $V_{i,Jan17}$, $V_{i,Feb7}$, and $V_{i,Nov7}$ indicate whether voter i voted in the January 17, February 7, and November 7, 2023 municipal elections, respectively. Each election indicator equals 1 if the voter had a recorded vote in that election and 0 otherwise. We define the outcome as:

$$\text{DidNotVote}_i = \begin{cases} 1, & \text{if } V_{i,Jan17} + V_{i,Feb7} + V_{i,Nov7} = 0 \\ 0, & \text{if } V_{i,Jan17} + V_{i,Feb7} + V_{i,Nov7} \geq 1 \end{cases}$$

In other words, voters were coded as `DidNotVote` = 1 if they did not vote in any of the three 2023 municipal elections, and `DidNotVote` = 0 if they voted in at least one of them. In the final dataset, approximately 71% of voters were classified as 2023 municipal non-voters.

The supplemental model used a broader engineered target variable: `low_propensity_municipal_adjusted`. This variable identifies voters whose age- and registration-adjusted municipal turnout rate was less than or equal to 25%. Unlike the primary target, which focuses only on 2023 municipal election behavior, this supplemental target captures a longer-run pattern of municipal election disengagement. The purpose of the supplemental model was not to replace the primary model, but to test whether the same reduced predictor set also generalized to a historical definition of low municipal participation.

To reduce leakage and improve interpretability, we removed variables that could directly reveal the outcome or duplicate voting behavior. These excluded fields included voter IDs, contact information, raw election-history columns, the target election fields themselves, vote-frequency summaries, voting-performance variables, and other direct turnout measures. This step was important because the goal was to predict municipal non-participation risk using demographic, household, socioeconomic, political, residential, and geographic characteristics rather than variables that directly encode whether someone voted.

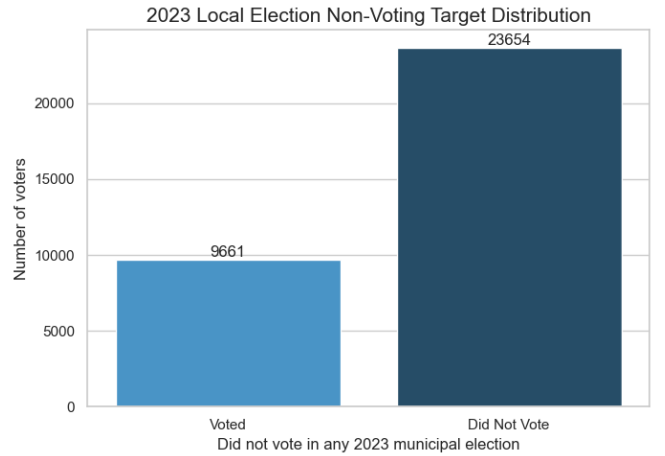


FIG. 7. **Distribution of the 2023 local-election non-voting target.** This figure shows the class balance for the 2023 municipal election turnout outcome in the cleaned voter file. Voters are grouped according to whether they voted or did not vote in the 2023 local election. The distribution shows that non-voters make up the majority of the target population, with 23,654 voters classified as not voting compared with 9,661 voters who participated.

The final reduced predictor set contained 18 variables. These predictors were selected because they are interpretable and connected to existing theories of voter turnout, including demographic characteristics, political affiliation, socioeconomic resources, household context, residential stability, and geography.

TABLE II. Categories of Predictors Used in Classification

Category	Variables Represented
Demographics	Age, gender, ethnicity, language
Political Context	Party affiliation, household political composition
Socioeconomic Status	Education, income, occupation, homeownership
Household Context	Family structure, household composition, residential stability
Geography	ZIP code, precinct, ward

Before modeling, numeric and categorical variables were processed separately. Numeric predictors were imputed using the median and standardized using `StandardScaler`. Categorical predictors were imputed using the most frequent category and transformed using one-hot encoding. This preprocessing was implemented through a `ColumnTransformer` pipeline so that all transformations were learned only from the training data and then applied consistently to the test data.

The data were split into training and testing sets using a 75/25 stratified train-test split with random state 141. Stratification preserved the same approximate target-class balance in both the training and test sets. The main model training set contained 24,986 observations, and the test set contained 8,329 observations. The target rate was approximately 71% in both the training and test sets.

We compared four classification models: Logistic Regression, Random Forest, Gradient Boosting, and XG-

Boost. Logistic Regression served as an interpretable baseline model, while the tree-based models were included because they can capture nonlinear relationships and interactions among predictors. XGBoost was selected as the final model for both the primary and supplemental analyses because it achieved the strongest overall performance while preserving the compact 18-variable predictor set.

In addition to the train-test evaluation, we used 3-fold stratified cross-validation on the training data. Cross-validation provided a more stable estimate of model performance by evaluating each model across multiple splits of the training set. This helped confirm that the selected XGBoost model was not performing well only because of one favorable train-test split.

After training the final XGBoost model, predicted probabilities were converted into four non-voting risk groups to make the model outputs more interpretable for outreach. Voters with predicted probabilities from 0.00 to 0.25 were labeled low risk, 0.25 to 0.50 moderate risk, 0.50 to 0.75 high risk, and 0.75 to 1.00 very high risk. These categories were used only to summarize and interpret predicted non-voting risk; model performance was still evaluated using the continuous predicted probabilities and standard classification metrics.

V. MODEL EVALUATION

Model performance was evaluated using accuracy, precision, recall, F1 score, and ROC AUC for `DidNotVote` = 1. Because the practical goal of the project is to identify voters likely to skip municipal elections, recall and F1 score were especially important. High recall means the model successfully captures a large share of actual non-voters, while F1 balances this recall against precision. ROC AUC was used to evaluate the model’s overall ability to rank higher-risk voters above lower-risk voters across probability thresholds.

We also tested models that included previous non-municipal election participation and voter age. These models performed substantially better, with the XGBoost model reaching a ROC AUC of approximately 0.92. This result suggests that prior voting behavior, especially participation in previous general, primary, or presidential primary elections, is extremely predictive of whether someone will participate in a municipal election. In other words, voters who have voted before tend to continue voting, while voters with little prior participation are more likely to skip future elections. However, because these voting-history variables are highly predictive and closely related to the outcome, they also create a risk of leakage and multicollinearity. For the main analysis, we therefore report the reduced model, which excludes direct voting-history measures and instead uses a more interpretable set of demographic, socioeconomic, household, political, residential, and geographic predictors.

Metrics

TABLE III. Model Classification Metrics

Model	Accuracy	Precision	Recall	F1	AUC
XGBoost	0.777	0.808	0.899	0.851	0.803
Random Forest	0.741	0.857	0.763	0.807	0.801
Gradient Boosting	0.774	0.805	0.900	0.850	0.800
Logistic Regression	0.720	0.856	0.728	0.787	0.786
Supplemental Model	0.728	0.741	0.897	0.812	0.729

XGBoost was selected as the final primary model because it achieved the highest ROC AUC among the reduced models while maintaining strong recall and F1 performance. Although Gradient Boosting produced a similar recall and F1 score, XGBoost provided the strongest overall ranking performance. For consistency with the primary analysis, we applied the same set of candidate models to the supplemental target and again selected XGBoost as the final classifier. This allowed the supplemental model to test whether the same 18 predictors and modeling approach generalized to a broader definition of municipal disengagement. The supplemental XGBoost model performed somewhat worse than the primary model by ROC AUC, but it retained a recall of 0.897 and a strong F1 score of 0.812. This suggests that the same 18 predictors are more effective for predicting 2023 non-voting specifically, but still capture broader patterns of low municipal participation.

The ROC curve in Figure 8 shows that the reduced candidate models performed better than random classification across probability thresholds. The reduced XGBoost model achieved the highest ROC AUC among these reduced models, approximately 0.803, indicating that it was the strongest reduced model at ranking likely 2023 municipal non-voters above voters. By contrast, the expanded XGBoost model that included previous non-municipal election participation and age achieved a much higher ROC AUC of approximately 0.92. This confirms that past voting behavior predicts future voting behavior very well. However, we opted to use the reduced model for the main analysis because it avoids overreliance on direct turnout-history variables and provides more actionable insight into the demographic, household, socioeconomic, and geographic factors associated with municipal non-participation.

The confusion matrix in Figure 9 provides a more direct view of how the final reduced XGBoost model classified voters at the selected threshold. The model correctly identified a large share of actual 2023 municipal non-voters, which is consistent with its high recall score of 0.899. The model also produced some false positives, meaning that some voters who did participate were classified as likely non-voters. For an outreach-oriented application, this tradeoff may be acceptable because the model is being used to prioritize outreach rather than make a punitive or high-stakes decision about individual voters.

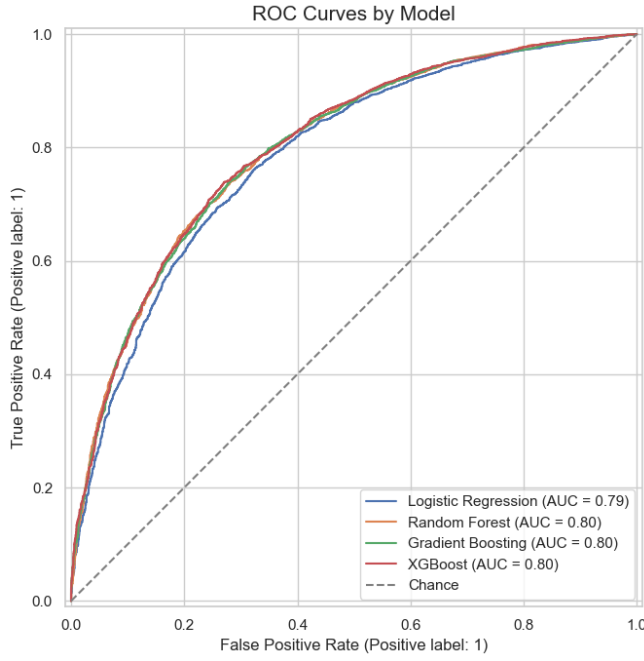


FIG. 8. **Model performance comparison using ROC curves.** This figure compares the predictive performance of the logistic regression, random forest, gradient boosting, and XGBoost models using receiver operating characteristic (ROC) curves. All models perform substantially better than random classification, shown by the dashed diagonal chance line. The tree-based models slightly outperform logistic regression, with random forest, gradient boosting, and XGBoost each achieving an AUC of approximately 0.80, compared with 0.79 for logistic regression.

VI. FINDINGS

The final reduced XGBoost model produced meaningful separation between voters with low and high predicted risk of municipal non-participation. After predicted probabilities were converted into four risk groups, actual non-voting rates increased steadily across the categories. This indicates that the model’s predicted probabilities were useful not only for classification, but also for ranking voters by relative outreach priority.

TABLE IV. Non-Voting Rates by Predicted Risk Group

Risk Group	Voters	Actual	Predicted
Low Non-Voting Risk	1,364	0.144	0.204
Moderate Non-Voting Risk	5,769	0.364	0.370
High Non-Voting Risk	7,525	0.630	0.640
Very High Non-Voting Risk	18,657	0.891	0.881

The model’s average predicted non-voting probabilities closely matched the actual non-voting rates. Only 14.4% of voters in the low-risk group failed to vote in the 2023 municipal election cycle, compared with 89.1% of voters in the very high-risk group. This suggests that the model can identify voters who are meaningfully more likely to

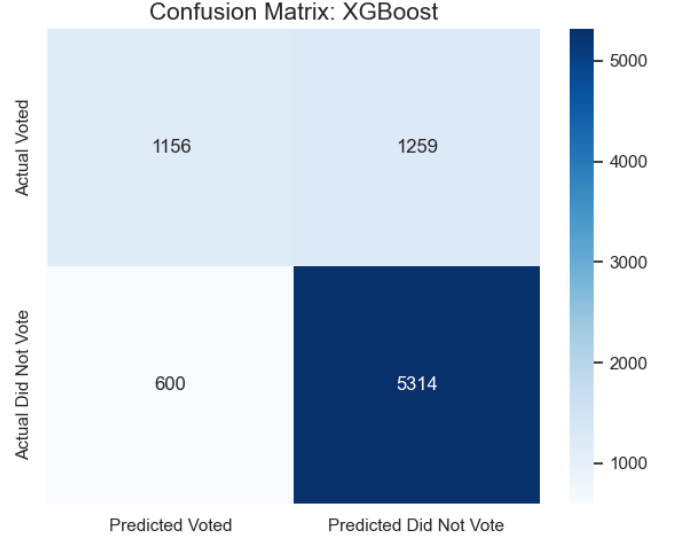


FIG. 9. **Confusion matrix for XGBoost non-voting classification.** This figure shows the XGBoost model’s predictions for whether voters participated or did not participate in the target municipal election. The model correctly identifies a large share of actual non-voters, with 5,314 true non-voters classified as “Did Not Vote,” while 600 actual non-voters were incorrectly classified as voters. Among actual voters, 1,156 were correctly classified as voters, while 1,259 were incorrectly classified as non-voters. Overall, the matrix suggests that the model is especially useful for identifying likely non-voters, which is central to the goal of targeting low-propensity municipal voters.

skip municipal elections, making the risk scores useful for outreach prioritization.

Permutation Importance

Permutation importance showed that the strongest predictors of municipal non-voting were age, modeled ethnicity and language-related grouping, homeownership probability, length of residence, income ranking, household political context, and precinct, shown in Figure 10. Age was the most important predictor by a substantial margin, suggesting that generational differences are central to municipal turnout patterns. Housing and residential stability variables were also important, indicating that voters with different levels of attachment to place may have different probabilities of participating in local elections.

One of the most important findings from the modeling process is that prior voting behavior was the strongest predictor of future voting behavior. In the full model, the most significant variables were related to whether a voter had participated in previous non-municipal elections, such as general elections and primaries. This confirms one of the central patterns from the EDA: voters who vote regularly tend to keep voting, and voters who

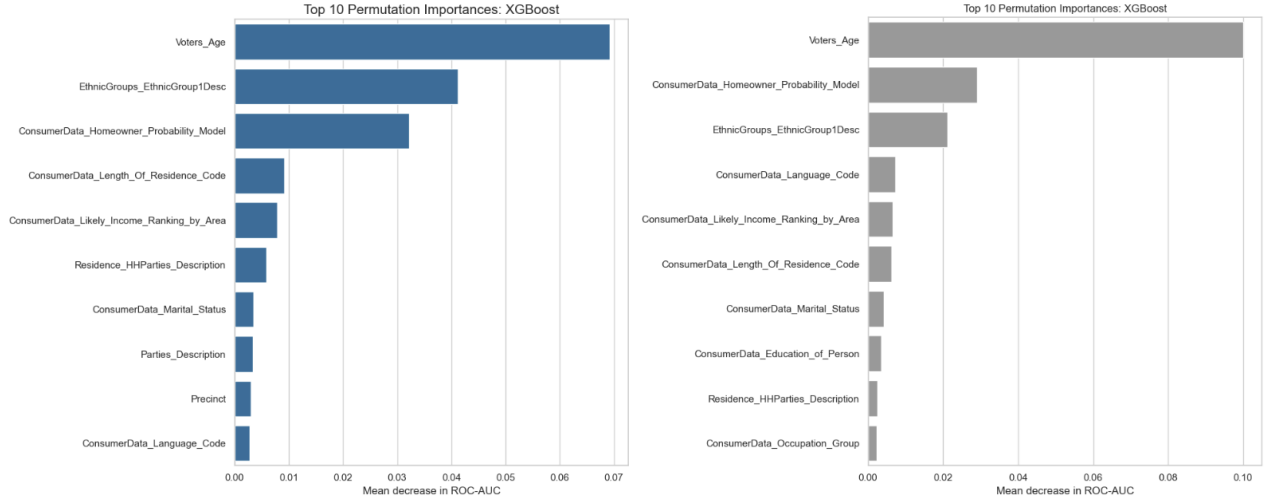


FIG. 10. **Top permutation importances for the primary and supplemental XGBoost models.** This figure reports the ten most important predictors from the primary and supplemental XGBoost models using permutation importance, measured as the mean decrease in ROC-AUC when each feature is randomly shuffled. The supplemental model uses **Low_turnout_municipal_rate**, an age-adjusted municipal turnout measure calculated as the number of municipal elections in which a voter participated divided by the number of municipal elections for which they were eligible. Across both specifications, voter age is the dominant predictor, while homeownership probability, ethnicity, language, income ranking, residence length, and party-related variables also contribute to predicting low municipal turnout propensity.

participate in municipal elections are more likely to participate in the next municipal election as well.

However, we did not keep these direct voting-history variables in the final reduced model because they were highly collinear with the outcome and risked creating leakage. If the model uses past election participation too directly, it may simply learn that “voters vote” and “non-voters do not vote,” rather than identifying broader demographic, geographic, household, and socioeconomic factors associated with municipal non-participation. For this reason, we removed direct turnout-history variables and instead used the final 18 selected predictors, which provided a more interpretable and actionable model for outreach planning.

This tradeoff is important. The full model may be slightly stronger for pure prediction, but the reduced model is more useful for understanding where outreach should be directed and why certain groups may be at higher risk of municipal non-participation. By focusing on the 18 selected predictors, the model avoids relying too heavily on circular voting-history information and instead highlights characteristics that can help guide non-partisan outreach before the 2027 mayoral election.

The permutation results for the main model show that age had the largest effect on model performance when randomly shuffled, indicating that age provided the strongest unique predictive signal. Ethnicity-related grouping, homeownership probability, length of residence, and income ranking also reduced model performance when disrupted, suggesting that municipal non-voting risk is connected to both demographic characteristics and residential or socioeconomic stability.

The supplemental model, which used the alternative low-propensity municipal turnout target, produced a similar pattern of variable importance shown in Figure 10. Age remained the strongest predictor, followed by homeownership probability, ethnicity-related grouping, language, income ranking, length of residence, marital status, education, household political context, and occupation. This consistency strengthens the interpretation that municipal non-participation is associated with generational differences, residential stability, socioeconomic context, and political attachment rather than a single isolated factor.

High-risk Groups

The high-risk profiles suggest that outreach efforts should prioritize younger voters, non-partisan voters, renters or voters with lower homeownership probability, and residents in higher-risk wards or precincts. Non-voting risk was especially concentrated among younger voters: 98.7% of voters aged 25–34, and 89.8% of voters aged 35–44 were classified as high or very high risk. Non-partisan voters also represented a major outreach group, making up approximately 69.0% of the high-risk population. Housing status showed another important divide, with renters having a high-risk rate of 95.8%, compared with 47.6% for homeowners. Finally, geography mattered: Ward 2 and Ward 4 showed especially elevated non-voting risk, suggesting that outreach should combine individual-level targeting with place-based mobilization.

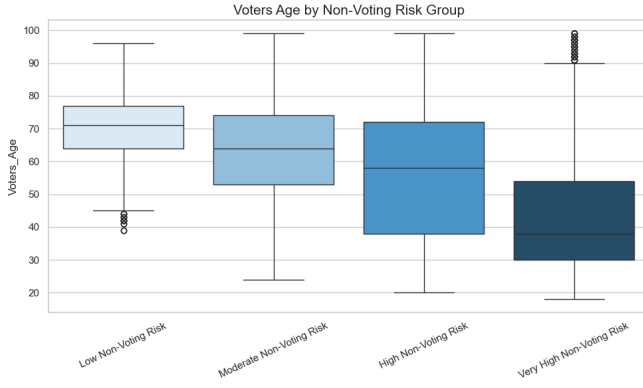


FIG. 11. **Predicted municipal non-voting risk by age group.** This figure compares the distribution of voter age across predicted non-voting risk categories. Voters classified as low non-voting risk tend to be older, while voters in the high and very high non-voting risk groups are generally younger. This pattern suggests that age is strongly associated with municipal turnout propensity, with younger voters more likely to be classified as at risk of non-participation.

VII. ADVICE AND RECOMMENDATIONS

For the 2027 mayoral election, Quincy should use our model as a starting point for nonpartisan voter outreach. The model is most useful as a prioritization tool: it helps identify voters and precincts where municipal non-participation risk is highest, but it does not explain every reason why a particular resident does or does not vote. Therefore, the model should be paired with local knowledge, community partnerships, and issue-specific outreach. Our framework can also be used for inference and outreach planning before an upcoming election. By estimating which voters are most likely to be low-propensity municipal voters, the model helps campaigns, civic organizations, or city partners identify where participation gaps are likely to be concentrated before Election Day.

Rather than treating the model as a final answer, the predicted risk scores should be used to tailor outreach strategies to different groups. For example, voters predicted to be low-propensity may need clearer information about why the election matters, where and how to vote, or how local offices affect issues such as schools, housing, public services, and representation.

1. **Prioritize voters with weak municipal voting history.** One of the clearest findings from the EDA and modeling process is that voting behavior is persistent. Voters who have participated in previous elections are generally more likely to vote again, and voters who have participated in municipal elections are more likely to participate in the next municipal election. In other words, if someone votes, they tend to keep voting; if someone repeatedly skips municipal elections, that pattern

can also persist. This supports prior research on turnout and habit formation, which suggests that voting can become a repeated civic behavior over time [6]. For Quincy, this means outreach should not only aim to increase turnout in 2027, but also help create repeat municipal voters.

2. **Focus on federal-only voters as a reachable group.** Federal-only voters are especially important because they have already demonstrated willingness to vote in higher-salience elections, but they do not consistently participate in municipal elections. Outreach to this group should not simply say “go vote.” Instead, it should explain why municipal elections matter for schools, housing, local taxes, public safety, infrastructure, and representation. These voters may already believe voting is important, but may not yet see city elections as connected to their daily lives.
3. **Use issue-based outreach, especially when local issues affect underrepresented communities.** The 2025 Quincy election context suggests that local issues can motivate participation when residents see a direct connection between the election and their community. As Dr. Wilcox noted, the debate over recognizing Lunar New Year as a school holiday may have helped mobilize Asian American voters, especially because Quincy has a large Asian American student population. This matters because the groups that consistently show up in municipal elections are often older, wealthier, and more white or European-origin voters. If issue-based mobilization can bring out communities that are usually underrepresented, then outreach for 2027 should connect the mayoral race to concrete local concerns rather than relying only on generic civic messaging.
4. **Segment outreach by voter type rather than using one message for everyone.** Younger voters, renters, newer residents, non-partisan voters, and federal-only voters may all have high municipal non-participation risk, but they may not be disengaged for the same reason. Younger or newer voters may need information about registration, polling places, and what city government controls. Renters and more mobile residents may be less connected to neighborhood political networks. Non-partisan voters may respond less to party-based appeals and more to candidate-specific or issue-specific messaging. Outreach should therefore be tailored to the group being contacted.
5. **Use precinct-level targeting and direct outreach.** The EDA showed that low-propensity municipal voting and federal-only voting are not evenly distributed across the represented Quincy precincts. Some precincts have much higher concentrations of voters who are politically engaged at

the federal level but inactive locally. Quincy should use these precinct patterns to prioritize outreach locations, such as schools, libraries, community centers, apartment complexes, workplaces, religious institutions, and commercial corridors. This aligns with experimental research on voter mobilization showing that direct contact and social-pressure-based outreach can increase turnout [7].

6. **Begin outreach early and repeat it over time.** Outreach should not begin only in the final weeks before the 2027 mayoral election. Research on youth civic engagement emphasizes that effective turnout work is early, repeated, and connected to civic learning and local relevance [4]. For Quincy, this could include voter guides, candidate forums, school-based outreach, multilingual informational materials, community meetings, and reminders that explain both how to vote and why the election matters.
7. **Use model risk groups as outreach tiers, not fixed labels.** The model should be used to prioritize outreach, not to make deterministic judgments about individual voters. High-risk and very-high-risk voters should receive more intensive outreach, but lower-risk voters should not be ignored. This is especially important because one problem in low-turnout elections is that campaigns often contact only habitual voters, which can reinforce existing participation gaps. A responsible use of the model should expand outreach to under-contacted voters rather than narrow the electorate further.
8. **Evaluate which outreach strategies actually work.** The model can identify voters who are at higher risk of skipping municipal elections, but it cannot prove which message or intervention will cause someone to vote. For the 2027 cycle, Quincy should track outreach efforts by precinct and voter segment, then compare turnout outcomes across different outreach methods. This could include nonpartisan mailers, text reminders, canvassing, school-based outreach, workplace outreach, and issue-based voter education. Over time, this would allow Quincy to move from prediction to evidence-based turnout intervention.

This project provides a framework for broadening participation in Quincy municipal elections. Its main contribution is not that it perfectly predicts turnout, but that it helps identify where participation gaps are concentrated and which groups may need more attention before the next mayoral election. Used carefully, the findings and model can help Quincy move from a general concern about low turnout toward a more targeted, data-informed, and locally grounded engagement strategy.

VIII. LIMITATIONS

Several limitations should be considered when interpreting these results. The supplied voter-file extract does not cover all 31 official Quincy precincts, so the geographic findings should not be treated as a complete city-wide assessment. Additional voter-file coverage would be necessary before applying precinct rankings across all of Quincy. We also did not have individual-level 2025 municipal turnout data available in the final modeling dataset, which is important because one of the motivating questions for the project was whether increased turnout in the 2025 local election could help inform outreach for the 2027 mayoral election. Without 2025 individual-level turnout records, we cannot directly model which voters newly turned out in 2025, which precincts improved relative to prior municipal cycles, or whether specific issues such as Lunar New Year mobilized particular communities at the individual level. If 2025 municipal voter history becomes available, the model should be updated and re-estimated.

Several important predictors, including ethnicity or language grouping, homeownership probability, income ranking, and other consumer characteristics, are vendor-modeled rather than directly observed or self-reported. These fields may contain measurement error, may have uneven accuracy across communities, and should not be interpreted as verified individual characteristics. They are more appropriate for evaluating broad patterns of representation and access at an aggregate level. The models are also predictive rather than causal. They identify characteristics associated with municipal non-voting, but they do not show why a person did not vote or which intervention would cause that person to participate. In particular, a high predicted risk score does not necessarily mean that a voter is persuadable. The model should therefore be used to prioritize outreach, not to make causal claims about individual behavior.

The primary outcome combines participation across three different 2023 municipal elections. These elections may have differed in visibility, competitiveness, and voter motivation. Coding a person as a voter if they participated in any one of the three elections creates a useful overall municipal participation outcome, but it may obscure differences between special and general municipal elections. Future work could model the November municipal election separately or compare outcomes across individual election dates. In addition, the model was trained using 2023 behavior and may not generalize perfectly to the 2027 mayoral election. Candidate competition, salient issues, campaign activity, election administration, and the composition of the electorate may change over time. The model should therefore be updated using newer individual-level turnout records when they become available, especially 2025 municipal turnout data.

Finally, the choice of probability threshold and outreach-risk categories affects who is labeled high risk. The categories are useful for prioritization, but they

should not be treated as natural or permanent divisions between voters. Performance, calibration, recall, and false-positive rates should also be evaluated separately across demographic and geographic groups before the scores are used operationally.

IX. CONCLUSION

This project shows that Quincy can approach the 2027 mayoral election with a more targeted, data-informed understanding of municipal turnout. Instead of treating low local participation as a general citywide problem, our analysis identifies where municipal non-participation is concentrated, which voter groups are most at risk of skipping local elections, and how outreach can be prioritized before Election Day. The central finding is that municipal turnout is uneven across precincts, party registration groups, age groups, and household contexts, meaning that a single generic turnout strategy is unlikely to reach all low-propensity voters effectively.

The modeling results show that a reduced XGBoost model can successfully rank voters by their risk of municipal non-participation while using a compact and interpretable set of predictors. Although the expanded model showed that prior voting behavior is the strongest predictor of future turnout, those variables were removed from the final model because they were highly collinear with the outcome and offered limited actionable insight. The final 18-predictor model therefore provides a more useful framework for outreach because it focuses on demographic, socioeconomic, household, political, residential, and geographic characteristics rather than simply confirming that past voters tend to vote again.

For the 2027 mayoral election, the most important implication is that Quincy should prioritize voters who are reachable but locally inactive. Federal-only voters are especially important because they have already shown that they are willing to participate in higher-salience elections, but they do not consistently vote in municipal contests. Outreach to these voters should emphasize why local elections matter for schools, housing, public safety, infrastructure, city services, and representation. The 2025 election context also suggests that issue-based mobilization can matter, especially when residents see a direct connection between the election and community concerns such as the Lunar New Year school holiday debate.

The project also highlights that predictive modeling should support, not replace, community-based democratic engagement. The model cannot prove why a person does not vote, and it cannot determine which message will cause someone to participate. However, it can help Quincy identify where participation gaps are likely to be most severe and where outreach resources may be most useful. Used responsibly, the model can help civic organizations, campaigns, and city partners move beyond contacting only habitual voters and instead reach groups

that are often underrepresented in municipal elections.

Ultimately, the value of this project is not that it perfectly predicts the 2027 electorate, but that it creates a practical framework for broadening participation in Quincy municipal democracy. By combining voter-file analysis, precinct-level patterns, local issue context, and model-based risk scores, Quincy can design outreach strategies that are more targeted, more inclusive, and more responsive to the communities most affected by local government decisions.

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